Divyansh Shukla

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Experience

Power Pulse India 2026 – Present
Machine Learning Engineer Intern *India*

- Built an end-to-end thermal object detection pipeline for advanced driver-assistance systems, training YOLOv8n on the FLIR ADAS v2 dataset over 6 road-safety classes (person, bicycle, car, motorcycle, bus, truck).
- Orchestrated resumable multi-hour GPU training runs on Modal with persistent volumes, secret management, and Weights & Biases logging, so interrupted runs restart from the last checkpoint rather than from scratch.
- Wrote a COCO-to-YOLO converter that symlinks rather than copies images, fuzzy-matching FLIR category aliases onto contiguous class IDs.
- Tuned the recipe for single-channel thermal input by disabling hue/saturation augmentation and raising classification loss weight against background confusion; evaluation reports per-class AP50 and AP50-95 against a fixed baseline (mAP50 0.59, mAP50-95 0.36).

Education & Certifications

Vellore Institute of Technology Expected 2027
B.Tech, Computer Science and Engineering — Artificial Intelligence & Machine Learning *CGPA: 9.03 / 10*

Certifications: Applied Accelerated AI (NPTEL); Deep Learning with Computer Vision (NPTEL); Deep Learning Specialization (Coursera)

Technical Skills

Languages: Python, C++, CUDA
ML Frameworks: PyTorch, PyTorch Lightning, Hugging Face Transformers, TRL, Ultralytics (YOLO), Stable-Baselines3, vLLM
Deep Learning: diffusion models, flow matching, transformers, attention mechanisms (RoPE, AdaLN-Zero), classifier-free guidance, object detection, reinforcement learning (PPO), vision-language models
LLMs & Agents: prompt engineering, structured output extraction, function calling, agentic workflows (planning, memory, tool use), retrieval-augmented search, LLM evaluation, Gemini API
Simulation & Robotics: MuJoCo, Gymnasium, OpenCV, Raspberry Pi, embedded/edge inference
Backend & Infrastructure: FastAPI, Django, Flask, Modal, Weights & Biases, Hydra, Git, Linux, NumPy, pandas

Projects

RoPE-DiT: Diffusion Transformer with 2D RoPE github.com/divyanshuklai/Custom2DRoPEDiT

- Implemented a ∼5M-parameter Diffusion Transformer from scratch: patchifier, timestep and class-label embedders, AdaLN-Zero conditioning, and a custom multi-head attention module.
- Designed a true 2D rotary positional embedding splitting the attention head dimension into row and column halves, rotating queries/keys by each patch's (height, width) coordinates to encode relative 2D distance rather than flattened 1D or absolute positions.
- Trained on CIFAR-10 with a rectified flow objective (MSE against the velocity field $v = x_1 - x_0$), sampling via a hand-written Euler ODE solver with classifier-free guidance.

Zero-Shot ICE RMA: Quadruped Locomotion RL github.com/divyanshuklai/zero-shot-ice-rma

- Built a MuJoCo environment for the Unitree Go1 quadruped (30-dim proprioceptive observations, 12-dim joint-position actions, fractal terrain for domain randomization) and implemented the Rapid Motor Adaptation bioenergetics reward with curriculum-scheduled penalty weights.
- Diagnosed a failed 42M-step PPO baseline, tracing exploding value loss and absent gait discovery to unnormalized observations, penalty dominance, and missing previous-action context.

Fashion Search: Conversational Product Discovery github.com/divyanshuklai/fashion_search

- Built a FastAPI chat app turning natural-language queries into product cards across 10 Indian e-commerce stores, chaining Tavily search, Gemini 2.5 Flash structured extraction, and BeautifulSoup scraping.
- Engineered a priority-based image and price extractor (JSON-LD → Open Graph tags → keyword-matched images), benchmarking it as the baseline for an in-progress agentic variant with planning, memory, and tool use.

Achievements

Won the Smart India Hackathon Internal (Intra-Campus) Round, Vellore Institute of Technology (2024).